Thermochemical Simulation of the Iris Phenolic Liners

Setup and Accepted Results

MIT Rocket Team — Solid Propulsion Simulations

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Exploratory study. Four configurations are considered: sustainer and booster, each with an aluminium or carbon-fibre composite casing. Heat flux, firing duration, grain port and measured recession are unavailable. The user authorized explicit surrogate inputs and sensitivity studies. This report separates native numerical checks from physical calibration. A pilot is not a demonstrated motor lifetime.

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1 Executive summary

No defensible real-motor burnthrough time has been established. The corrected midpoint model has 4 audited terminal results among the four primary cases at this snapshot. Every history ends at its accepted horizon or property-scope stop; no later values are inferred. Six native prescribed-temperature checks verify the rebuilt material library. Historical runs exposed separate restart, internal-substep accounting, nonlinear-tolerance and reaction-temperature issues, documented in the verification sections. Their curves are not used as corrected-model predictions. Booster Al completed the prescribed seven-second heating pulse and cooling through 300 s. Its retained mesh thickness is 6.2706 mm, with a sampled inner-casing maximum of 84.78 degrees Celsius at 142.75 s. Booster CF completed the prescribed seven-second heating pulse and cooling through 300 s. Its retained mesh thickness is 6.2706 mm, with a sampled inner-casing maximum of 114.24 degrees Celsius at 102.75 s. Sustainer Al completed the prescribed seven-second heating pulse and cooling through 127 s. Its retained mesh thickness is 4.7030 mm, with a sampled inner-casing maximum of 120.50 degrees Celsius at 79.00 s. Sustainer CF stopped at its provisional casing-property flag near 34.000 s with 4.7030 mm of retained mesh; its later thermal history is outside this calculation's scope. Independent tests also show that the residual-solid cutoff materially affects sustainer CF scope timing. Current native values remain conditional on the sealed 0.1-percent removal rule; tighter-cutoff convergence is investigated separately. No native exhaustion event has been accepted in these scenarios.

The initial twenty-row phenolic mesh is insufficient for recession. Independent refinement and 54 input-sensitivity cases show that missing thermal inputs and the partial-cell removal approximation matter. Under sustained nominal heating, provisional casing-property flags are reached with substantial liner remaining. Those stopping points limit this model's scope; they are not motor lifetimes or structural failure predictions. The immediate objective is to complete the native refined histories and compare their temperatures, mass loss and cooling against independent calculations.

2 Purpose and definition of burnthrough

The primary event is exhaustion of the continuous phenolic barrier, leaving the casing exposed in this two-layer geometry. Pyrolysis conversion alone is not this event: char remains solid and can continue insulating. We report modelled solid mass, retained mesh thickness, whole-cell recession, conversion depth and casing temperatures separately. Retained mesh thickness is not a measurement of physical liner thickness. A numerically accepted exhaustion event would be bracketed by the last intact state and first exhausted state.

The calculation represents an initially intact, locally uniform wall. Cracks, joints, debonding, char spallation and particle erosion are not resolved. Material-temperature flags indicate the scope of surrogate properties, not certified pressure-casing failure. If the event is absent, the result is "not reached through the simulated exposure," with residual thickness, rather than an extrapolated lifetime.

3 Geometry and discretization

The supplied propellant radii are provisionally used as liner inner radii. No unreported gap or adhesive layer is inserted. Shared interface nodes impose perfect thermal contact. Actual bond/contact resistance and manufacturing thickness tolerances are unavailable and are not quantified by the material-property perturbations. The user requested short axial and sector slices with about twenty radial cells per material. Each exchanged STEP contains exactly two solids and is reimported independently to verify volumes and dimensions.

Motor	Bore radius	Phenolic	Case	Outside radius
	Dimensions in millimetres			
Sustainer	39.6875	4.7625	3.1750	47.6250
Booster	65.0875	6.3500	4.7625	76.2000

Al and CF versions share the dimensions within each motor. The initial domain is 0.5 mm long and 0.5° wide, with 2 axial and 2 angular divisions. Twenty radial cells in each material give 160 SOLID278 elements and 369 nodes. The axis is global Y. Lateral and axial faces are adiabatic under the uniform-wall assumption. Integrated mass/power scales by 720 to a full ring of the same short length; temperatures and recession do not scale.

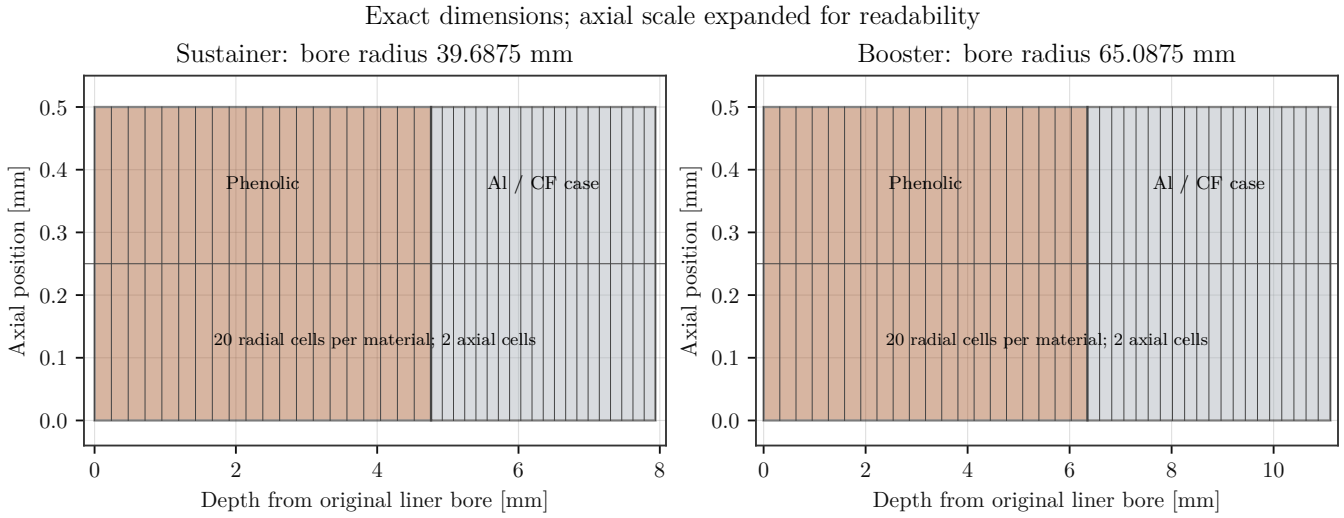


Figure 1. Radial/axial mesh projections. The angular direction has two divisions; axial scale is expanded. These are setup diagrams, not solved contours.

STEP is the exchange geometry. The native structured mesh is generated deterministically from the same dimensions; it is not an automatic mesh imported from STEP. The initial 20-cell phenolic pitches are 0.238125 mm and 0.3175 mm. Independent comparisons at 40, 80 and 160 cells show sensitivity to thermal gradients and cell-removal quantization. Refined native inputs use 80 phenolic and 20 casing rows, giving 400 elements and 909 nodes. Further independent refinement assesses retained partial-cell storage and time accuracy.

4 Materials and thermal environment

4.1 Phenolic and aluminium inherited from the earlier study

The reference virgin and char densities are 1250 and 600 $\mathrm{kg\,m^{-3}}$. The inherited single-step kinetics use $A = 333\,\mathrm{s^{-1}}$, $E_a = 64\,081\,\mathrm{J\,mol^{-1}}$, order one, and an endothermic pyrolysis term of $418\,\mathrm{kJ\,kg^{-1}}$. Thermal tables vary with temperature. Their applicability to the manufactured Iris liner is uncalibrated. Aluminium initially uses $\rho = 2700$, $k = 167$ and $c_p = 896$ in SI units; the exact alloy is unspecified.

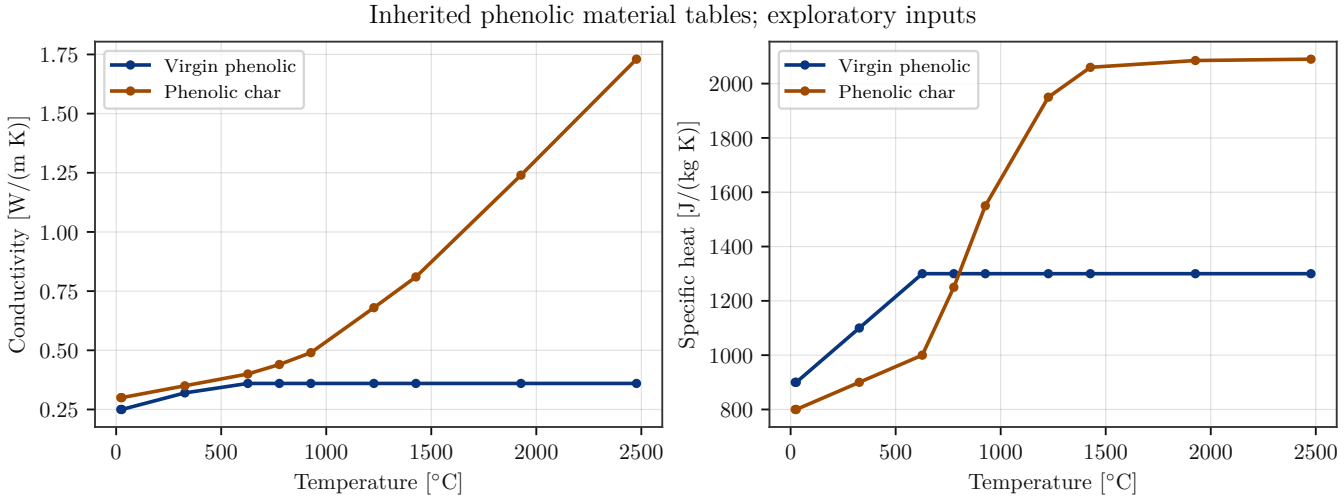


Figure 2. Inherited input tables. Lines interpolate prescribed material data and are not experimental measurements made for Iris.

4.2 Carbon fibre surrogate

The user's ACP fin tutorial supplies $\rho = 1451\,\mathrm{kg\,m^{-3}}$, elastic and thermal-expansion data, but no CF conductivity or heat capacity. Its G10 thermal values are not used as CF properties. The initial woven-laminate surrogate has radial

conductivity $0.55\,\mathrm{W\,m^{-1}\,K^{-1}}$ and equal tangential/axial conductivities $2.5\,\mathrm{W\,m^{-1}\,K^{-1}}$, within the woven-composite ranges reported by Joven and colleagues [3]. Local material frames follow the cylinder.

For heat capacity, use the separately identified T700/M21 surrogate $c_p = 2.8773T + 687.31$, with Celsius temperature and SI heat-capacity units [4]. Clamp this input above 200 °C; high-temperature predictions then have a flagged material approximation. The density, conductivity and heat capacity come from different references and do not constitute a measured property set for the user's casing. A 150 °C CF flag and 300 °C Al flag track provisional property scope, not structural failure. NASA's alternative carbon-epoxy data provide another sensitivity reference [5].

4.3 Imposed exposure

The initial gas-side assumption is $T_g = 2504\,\mathrm{K}$, unblown $h_0 = 1000\,\mathrm{W\,m^{-2}\,K^{-1}}$, and nominal pressure 800 psi, following the earlier study and the user's pressure estimate. Pressure is interpreted as absolute in the char-law closure. Initial and external ambient temperature is 22 °C; external convection is $8\,\mathrm{W\,m^{-2}\,K^{-1}}$ and emissivity is 0.25. The initial hot-radiation emissivity is zero, with separate radiation sensitivity scenarios.

The liner is assumed exposed from the start of imposed heating. The actual grain shielding and local exposure history are unknown. The campaign includes native pilots and 7 s baselines, refined native 7 s pulses with cooling, independent 7/20/60 s pulse screens, and a separate independent continuous-exposure branch. None of these pulse lengths is an inferred motor burn duration. Sensitivities vary heat-transfer coefficient, radiation, material properties, pressure and char-consumption parameters; they are not a statistical uncertainty envelope.

4.4 Supplied reaction and inventory envelope

The user's reaction is atom-balanced:

$$3\mathrm{NH_4ClO_4} + 3\mathrm{Al} \longrightarrow \mathrm{Al_2O_3} + \mathrm{AlCl_3} + 3\mathrm{NO} + 6\mathrm{H_2O}.$$

Its reactant mass bookkeeping is approximately 81.32 percent ammonium perchlorate and 18.68 percent aluminium. With constituent densities of 1950 and $2700\,\mathrm{kg\,m^{-3}}$, an ideal additive-volume binary mixture would have density $2056.7\,\mathrm{kg\,m^{-3}}$. The AP density is a published material value [6]; the binary mixture is an explicit assumption, not a measured propellant formulation.

Motor	Volume at 1 m [litres]	Full-cylinder mass [kg]	0.75–1.25 m range [kg]
Sustainer	4.948	10.18	7.63–12.72
Booster	13.309	27.37	20.53–34.22

These are full-cylinder envelope values with no port, binder or voids. Actual mass scales with occupied volume fraction and actual bulk density. The stoichiometric product list is not an equilibrium combustion calculation, and this inventory cannot establish burn duration, wall heat flux or char kinetics. The inherited gas temperature remains an imposed scenario. No full-length fuel cylinder is meshed.

5 Thermochemical physics and solver

The phenolic thermal law solves conduction with temperature/conversion-dependent properties, endothermic pyrolysis and gas sensible cooling. The reaction state follows

$$\dot{\alpha} = A\exp[-E_a/(RT)](1-\alpha), \qquad 0 \le \alpha \le 1.$$

The intended reaction calculation integrates conversion analytically at the temporal midpoint, $(T_{n+1} + T_n)/2$, and requests a smaller solver increment when conversion changes too much. The inherited native routine actually used $T_{n+1} + (T_{n+1} - T_n)/2$: ANSYS supplies the current endpoint temperature, not the starting value. The corrected expression subtracts half the increment. The same correction applies to gas sensible cooling, which uses that temperature. The native diagnostic evidence and correction status are reported below. The gas source follows the virgin-to-char density loss. A surface blowing correction reduces convection as gas leaves the solid. Pore pressure and Darcy transport are not resolved.

A separate effective char-consumption law combines kinetic and mass-transfer limits and available surface energy. The inherited effective removal cost is $35\,\mathrm{MJ\,kg^{-1}}$. This is an exploratory closure, not a calibrated oxidation mechanism. A row is scheduled for removal only after residual solid mass falls below 0.1 percent of its initial virgin mass. Hot-face loads move to the next live row. Consumed char remains in the live finite-element thermal representation until row removal; its numerical influence must be bounded through refinement or a revised partial-cell treatment.

MAPDL 2026 R1.02 runs natively in Windows through the Parallels VM, using a versioned x64 material DLL and the installed Python coupling controller. The initial tiny models use one process and a row-summed (diagonal)

specific-heat matrix, matching the SOLID278 default without advection. At every coupling step, the controller checks exported time and live element identity, gas/char/solid mass, thermal energy and the actual applied hot-face power before accepting an atomic checkpoint. Native face heat rates are used, with independent reconstruction of user-material energy sinks where the native aggregate channels omit them.

The initial gates are mass residual at most 10^{-6} of initial liner mass, step energy residual at most 5 percent, and hot-load error at most 2 percent. These numerical gates do not validate the material law. Comparison of mesh and coupling steps targets less than 5 percent change in event time, with event brackets and temperatures reported explicitly.

The corrected pilot exporter retrieves only defined live-material ranges. The earlier three undefined-SVAR warnings per step were removed without changing physical equations. The native user-material solve retains the standard warning that the Ansys-supplied USERCK routine is used; its banner separately confirms a user-defined thermal material. This known warning is recorded, not silently counted as zero warnings. Longer runs also report elapsed-time/CPU-time performance warnings; these are retained in the validation records.

Both corrected CF primary runs also record a material-table warning at 21.9999998 degrees Celsius, approximately 2×10^{-7} K below the initial 22-degree table endpoint. The casing conductivity is constant in that interval. This negligible numerical undershoot is retained in the audit; it is distinct from the controller's explicit 150-degree material-scope stop. ANSYS ceased its own repeated temperature-range checking after the warning, while the independent controller scope checks continued at every accepted interval.

6 Execution and storage controls

The running VM has 14 virtual CPUs and approximately 112 GiB RAM. The repository is accessible through a dedicated I: mapping. A native no-solve probe completed with a successful license checkout and zero warnings/errors. Each case receives a separate sealed runtime; unchanged source hashes and expected inventories are checked before launching.

The user's storage floors are 400 GB decimal free on the Mac disk and 10 GB free on Windows C:. A host watcher checks the Mac every five seconds and the guest disk about every thirty seconds. The controller also requires a fresh storage heartbeat. On a threshold crossing, new work stops, a stop latch is written and the active solver is asked to terminate cleanly. Interrupted states are not published as accepted data.

Native results/restarts remain in ignored storage. Major setup and report milestones are pushed to GitHub after verification; staged files and new Git objects at or above 100 MB are rejected. Reports freeze accepted state indices and source hashes. Recovery preserves the failed run and matches native restart state with controller state before continuation.

7 Accepted results at this edition

This snapshot was frozen at November 04, 2026 at 03:55 UTC-05:00. The table and primary histories use corrected midpoint cases only, with tighter heat-flow convergence and strict single-substep accounting. Historical results remain in their explicitly labelled verification sections. Longer runs are displayed only as their states are accepted; queued cases do not acquire predicted ANSYS values. These results are provisional until spatial and temporal comparisons are complete. No burnthrough event has been accepted.

Run	Time [s]	Hot face [°C]	Remaining [mm]	Energy error [%]
Booster Al (n=80, midpoint)	300.0000	72.59	6.27062	0.0002
Booster CF (n=80, midpoint)	300.0000	91.49	6.27062	0.0004
Sustainer Al (n=80, midpoint)	127.0000	110.61	4.70297	0.0002
Sustainer CF (n=80, midpoint)	34.0000	249.54	4.70297	0.0012

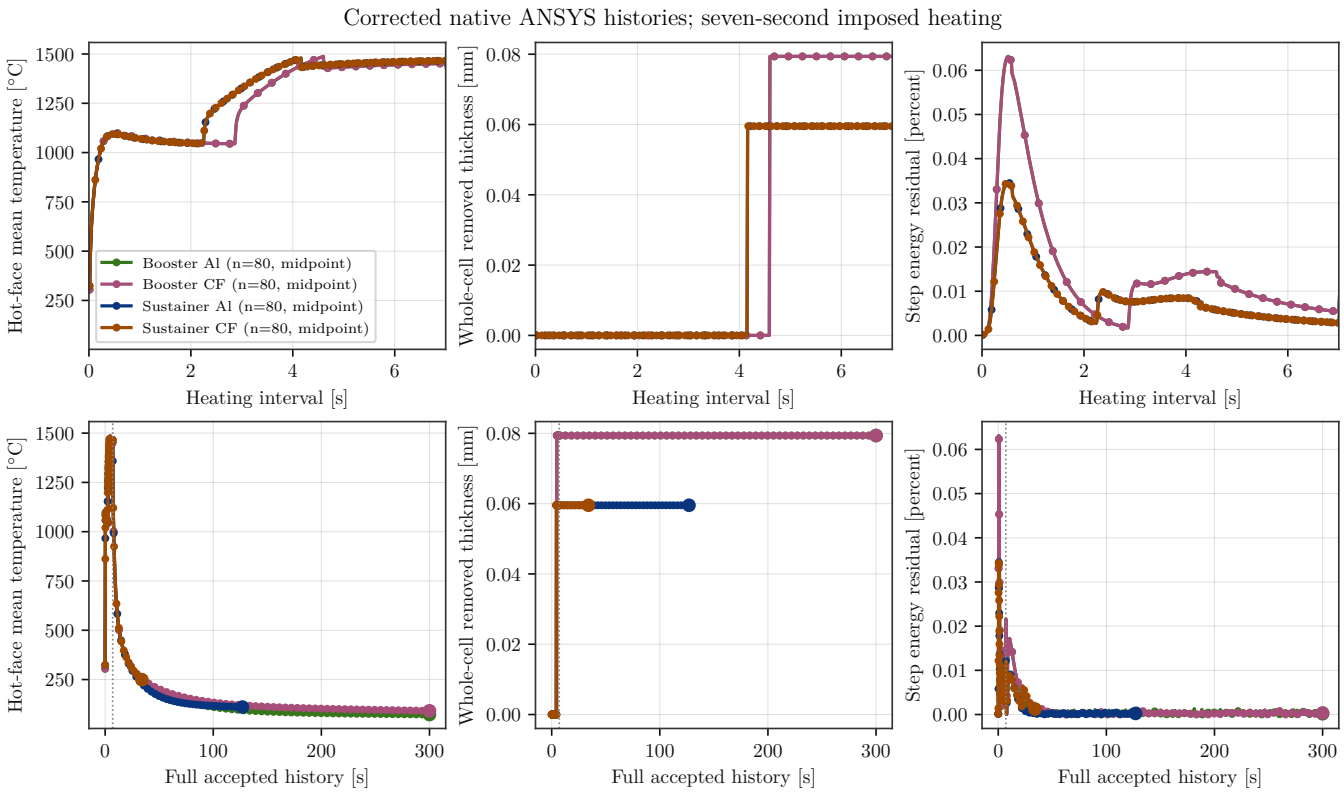


Figure 3. Accepted native histories. No extrapolated future states are included.

8 Verification findings and current limitations

The native steady-conduction benchmark used the same sector mesh, convection face numbers and CF material frames with constant properties and reactions disabled. All four configurations matched the analytical two-layer cylinder within 0.013 K; the acceptance target was 0.1 K. This checks geometry, thermal resistance and orthotropic axes, not the reaction law. The deliberately artificial steady temperatures in that benchmark are not motor predictions.

Manufactured storage faults verified both free-space floors, the latched stop and the correct native abort filenames. A communication fault is retried as a read-only query; it is not reported as a disk threshold crossing. A guest-local pilot reproduced every shared-drive pilot nodal temperature exactly. It offers an alternate execution location while mirroring accepted results to the Mac. The controlled checkpoint restart matched an uninterrupted native run at 1.1 s, including conversion state and energy integrals. The maximum nodal temperature difference was 0 K.

The independent cylindrical finite-element model uses a separate radial assembly, two Gauss points per cell, backward Euler time integration and row-summed heat capacity. It uses the intended midpoint reaction temperature. Early comparisons with the inherited native routine differed by approximately 0.70 K at 0.1 s and 0.97 K at 1.0 s, but did not expose its temperature-increment convention error. The later diagnostic checks below address that distinction. Numerical agreement is not physical calibration.

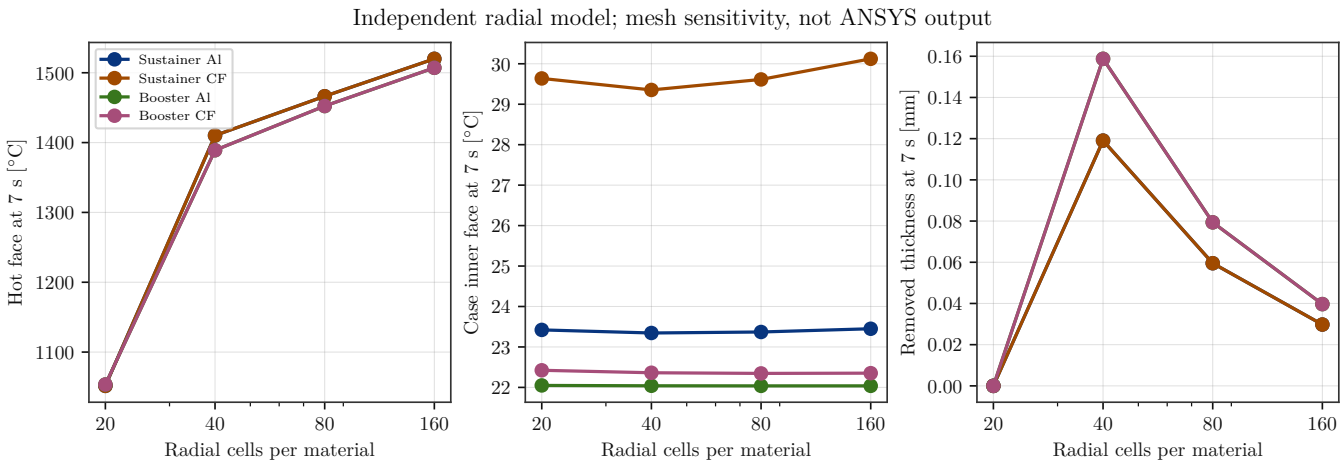


Figure 4. Independent seven-second radial mesh checks at a 0.0125 s coupling step. Coincident Al/CF hot-face and recession curves overlap. A cell-removal jump is not a smooth estimate of recession.

The twenty-cell mesh does not resolve recession sufficiently. The independent sustainer hot face at 7 s changes from about 1052 to 1410, 1466 and 1520 degrees Celsius at 20, 40, 80 and 160 radial cells. Its corresponding whole-cell removed thickness is 0, 0.1191, 0.0595 and 0.0298 mm. Case temperatures are much less sensitive over this short exposure. These comparisons justify refinement; they do not demonstrate a converged burnthrough time. Halving the 80-cell time step also changes hot-face temperature, so mesh and time accuracy remain separate checks.

Numerical long-exposure stress tests were also performed to expose weaknesses in the inherited removal scheme. They showed substantial mesh/time-step dependence, while the casing had already left its provisional property range. Those apparent exhaustion times are explicitly rejected as motor-lifetime results. The retained partial-cell thermal mass and lagged surface coupling require further assessment. Physically interpreted continuous-exposure runs now stop at the first casing property-range crossing. That crossing indicates a limit of this model, not an experimentally established structural failure.

9 Casing temperatures in accepted native states

The maxima below apply only through each accepted time shown. The listed times locate sampled maxima within that history; a material-scope stop cannot locate a later thermal peak. Inner and outer casing temperatures are kept separate because thermal resistance can delay outside heating while the liner-facing side becomes hotter.

Case	Through [s]	Inner max [°C]	At [s]	Outer max [°C]	At [s]
Booster Al (n=80, midpoint)	300.000	84.784	142.75	84.765	142.75
Booster CF (n=80, midpoint)	300.000	114.243	102.75	105.131	144.25
Sustainer Al (n=80, midpoint)	127.000	120.497	79.00	120.477	79.25
Sustainer CF (n=80, midpoint)	34.000	150.063	34.00	112.879	34.00

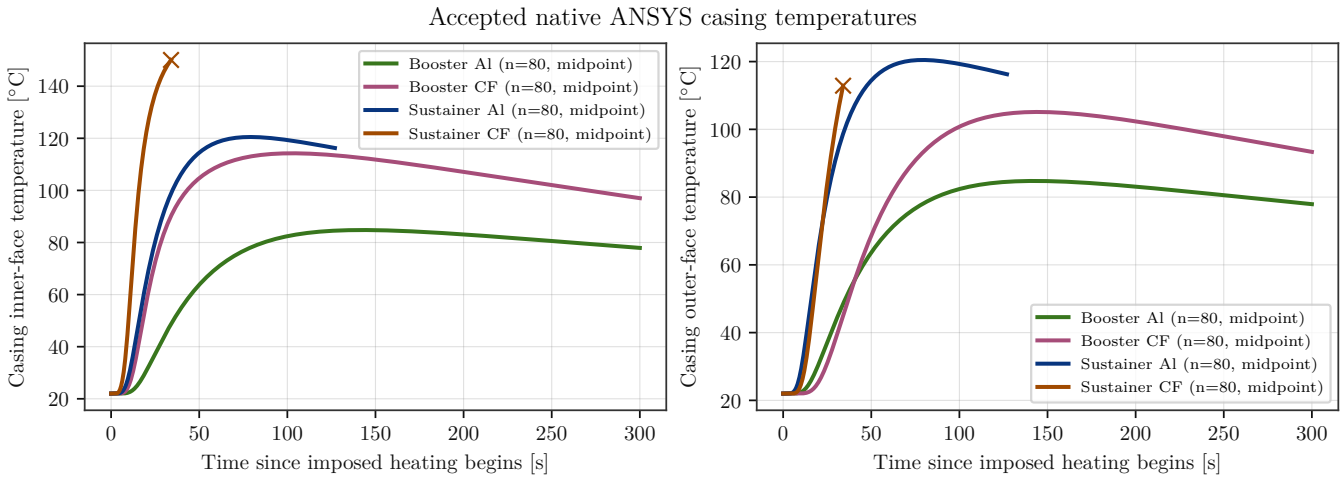


Figure 5. Native accepted casing-temperature histories. An x marks a provisional material-scope stop, not predicted failure. These are thermal calculations, not pressure-casing strength or qualification results.

9.1 Mass bookkeeping and retained partial cells

The next table uses fractions of the initial liner mass at each run's final accepted time above. Gas and consumed char are separate cumulative losses. The last column measures consumed char that remains represented thermally until its cell is removed; it is a numerical storage discrepancy, not extra physical material.

Case	Solid left [%]	Gas loss [%]	Char loss [%]	Retained in FE [%]
Booster Al (n=80, midpoint)	89.404	9.854	0.741	0.169
Booster CF (n=80, midpoint)	89.385	9.873	0.741	0.169
Sustainer Al (n=80, midpoint)	85.767	13.417	0.815	0.249
Sustainer CF (n=80, midpoint)	85.727	13.457	0.815	0.249

A zero whole-cell recession does not mean zero mass loss. Conversely, pyrolysis conversion alone is not a hole through the liner. Refinement is needed to bound the effect of the retained partial-cell storage on temperature and recession.

9.2 Accumulated numerical balance checks

The signed residual below checks accumulated bias; the absolute residual sum avoids cancellation of positive and negative step errors. Both are normalized by the time integral of the step balance scale. The independent mass check recomputes remaining solid plus gas, consumed char and expelled residual against the initial geometric inventory.

Case	Signed energy [%]	Absolute energy [%]	Mass relative error
Booster Al (n=80, midpoint)	0.0134	0.0142	3.2e-15
Booster CF (n=80, midpoint)	0.0132	0.0140	2.9e-15
Sustainer Al (n=80, midpoint)	0.0084	0.0088	3.3e-15
Sustainer CF (n=80, midpoint)	0.0086	0.0091	3.3e-15

The scale is $\max(|\dot{U}_{\mathrm{FE}}|, |Q_{\mathrm{conv}}|, |Q_{\mathrm{body}}| + |Q_{\mathrm{nodal}}|, 10^{-6}\,\mathrm{W})$ integrated over time. These residuals check the solved equations. They are not an independent physical enthalpy balance for departing material. Energy assigned to removed cells remains a separate diagnostic because the inherited capacity law is not a conservative enthalpy remap.

9.3 Corrected native temperature fields

Each map uses accepted corrected-midpoint ANSYS checkpoints only. The left panel resolves the imposed seven-second heating interval; the right panel shows the full accepted history, including heat soak and cooling. The dashed line marks the casing interface, while cyan marks the depth already removed during the solve. Inactive nodes are masked. Temperatures are radial-station averages of the small uniform sector; these are computed fields, not measurements.

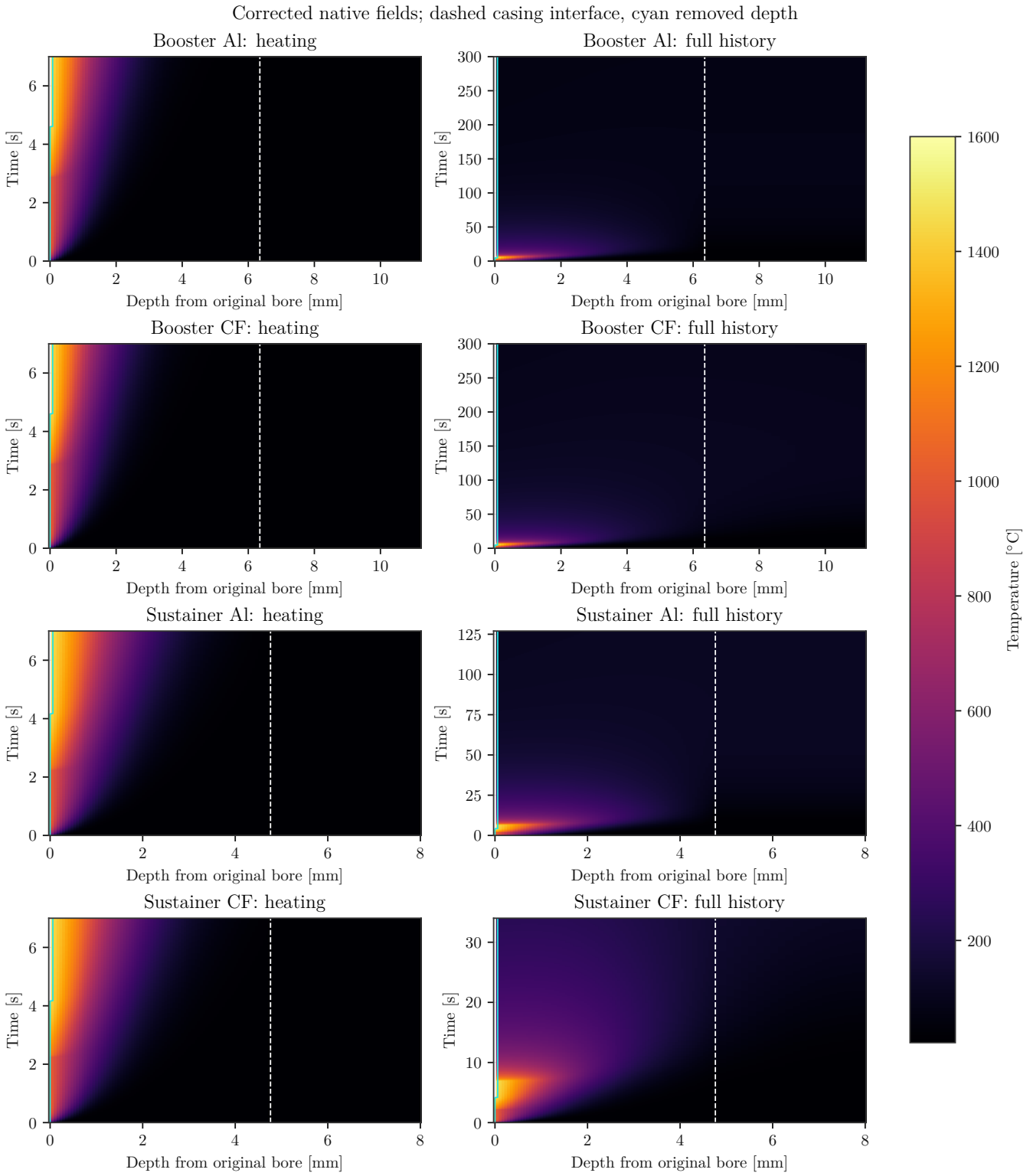


Figure 6. Corrected native temperature fields with a common scale. No time beyond an accepted checkpoint is shown.

9.4 Native temperature fields over the baseline exposure

These historical maps use every accepted checkpoint of the four original twenty-row native baselines, which used the inherited reaction-temperature expression. They are not predictions of the corrected midpoint model. Temperatures are averaged over each radial station of the uniform sector and interpolated for display. The dashed line marks the liner–casing interface. The cyan line is the upper-cell edge of the contiguous 98-percent conversion front; it is neither

recession nor burnthrough. All four coarse baselines retain every liner row through seven seconds, despite nonzero gas and char losses.

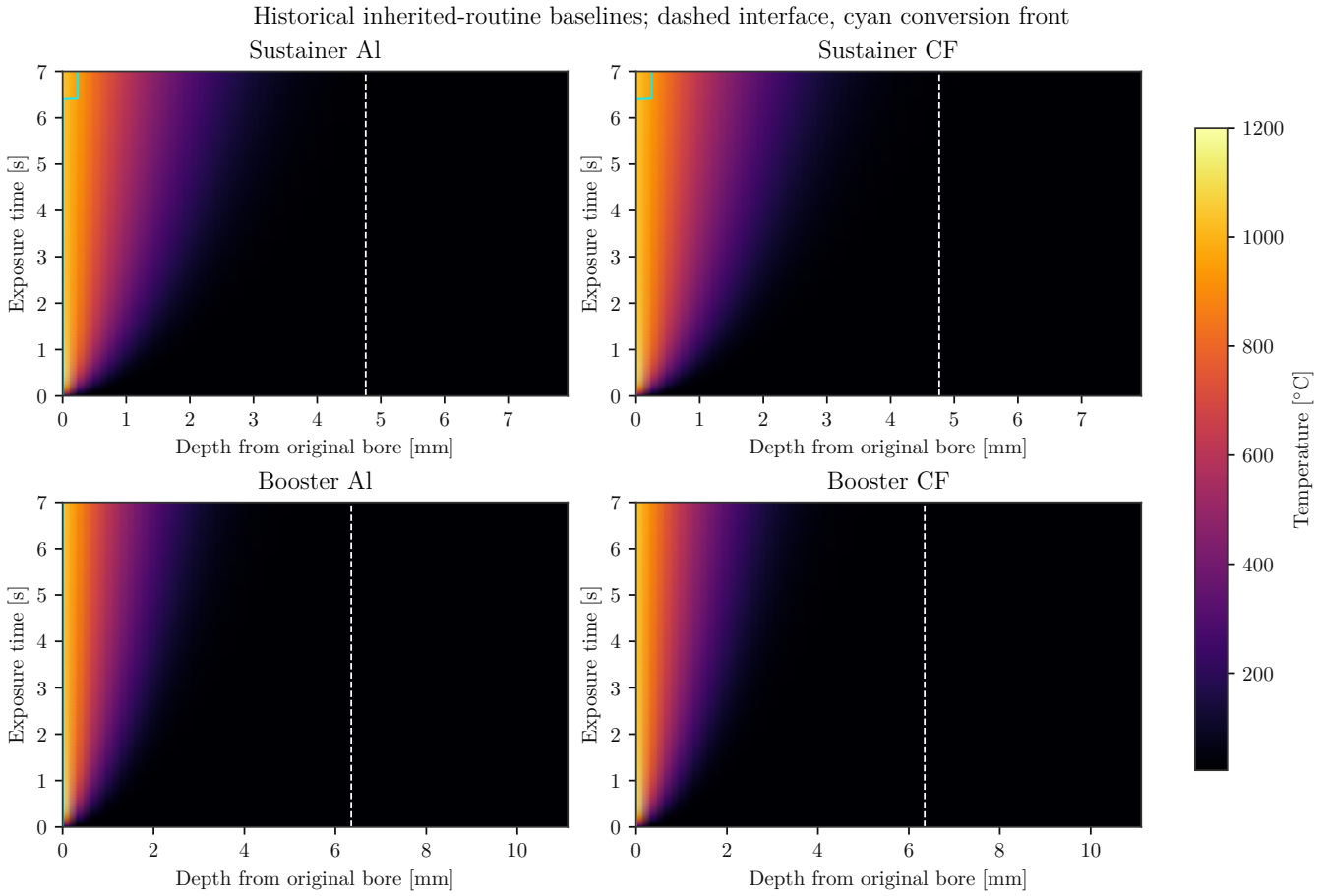


Figure 7. Native ANSYS baseline temperature fields, with a common temperature scale. These coarse-mesh fields are accepted numerical states; the recession discretization still requires the refined comparisons.

10 Independent numerical checks and scope limits

The following historical checks compare a separate cylindrical finite-element assembly with the original coarse ANSYS profiles at 0.1, 1, 2, 5 and 7 seconds. These native baselines used the inherited reaction-temperature expression. Passing the coarse-mesh comparison did not expose that defect; these records remain historical verification evidence, not corrected-model predictions. Numerical agreement cannot establish physical accuracy.

Case	Maximum profile difference [K]	Maximum conversion difference
Sustainer Al	1.0159	0.00263
Sustainer CF	1.0159	0.00263
Booster Al	1.0346	0.00246
Booster CF	1.0346	0.00246

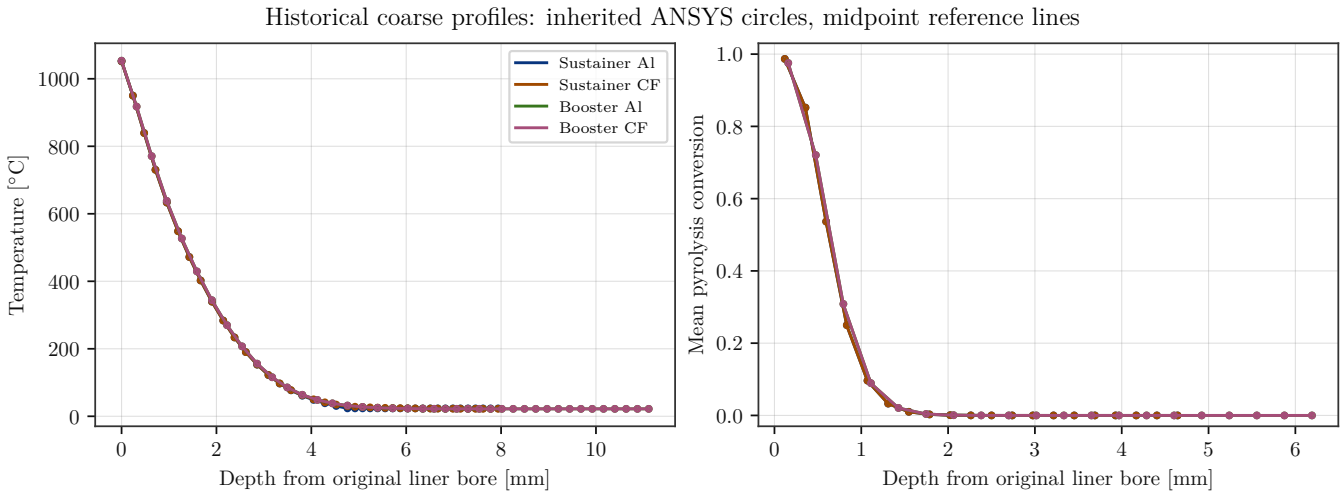


Figure 8. Accepted coarse-mesh native profiles and independent calculations. Temperatures are averaged over each uniform radial station. Agreement on the same mesh is not a mesh-convergence test.

10.1 Refined continuous-exposure stopping points

Under the nominal constant-heating assumptions, the independent model reaches the provisional casing property-range flag while a substantial liner remains. The table uses 320 cells per material and a 0.00078125 s step ceiling. These are model-scope stopping times, not burnthrough times, material failure temperatures or approved operating durations.

Case	Property flag [°C]	Scope ends near [s]	Retained mesh [mm]
Sustainer Al	300	49.485	3.0361
Sustainer CF	150	18.344	4.4053
Booster Al	300	87.274	2.9170
Booster CF	150	32.708	5.4570

The 160-to-320-cell joint refinement changes these stopping times by less than one percent. Separate 80-cell time-step comparisons change them by at most 3.6 percent. Remaining-thickness differences between the finest spatial cases are at most 0.020 mm. These numerical checks concern the property-scope event; they do not validate a later exhaustion event.

10.2 Native layer-removal verification

Sixteen manufactured steady states cover all four configurations with zero rows removed, the first row removed, only the final phenolic row remaining, and the entire phenolic removed. Exact live-element counts, moving hot-face powers and analytical cylinder temperatures passed; the largest temperature discrepancy was 0.00047 K. The final stage applies heating to the exposed casing. These are controlled software checks, not a calculated motor recession history.

The first static fixture omitted the required full Newton option and was rejected by MAPDL. The corrected fixture passed; production transient inputs already contained that option. Static load-factor warnings and unused zero outputs for dead-element heat-table entries are retained in the native log. A separate controller fixture verified one-time residual-mass ejection and termination at the final row, without indexing a nonexistent next liner row.

10.3 Input sensitivities within the provisional material scope

Each bar changes one input at 80 radial cells per material and a 0.003125 s step ceiling. The reference is the same-resolution nominal independent calculation. This is a sensitivity screen, not a probability distribution or a combined worst case. The twenty-percent phenolic perturbations and factor-two removal-cost range are engineering choices, not measured confidence bounds.

Pressure changes affect only the assumed char-reaction closure here; imposed gas temperature and heat-transfer coefficient remain fixed. A small pressure effect in this model does not imply that real motor pressure is unimportant. The NASA alternative uses its slide-14 carbon-epoxy thermal-model tuple: density 1596, conductivity 0.3832 and specific heat 886 in SI units, with a resistance model based on 39 percent resin and 61 percent fibre by volume [5]. It is a different composite, not an identification of the Iris casing.

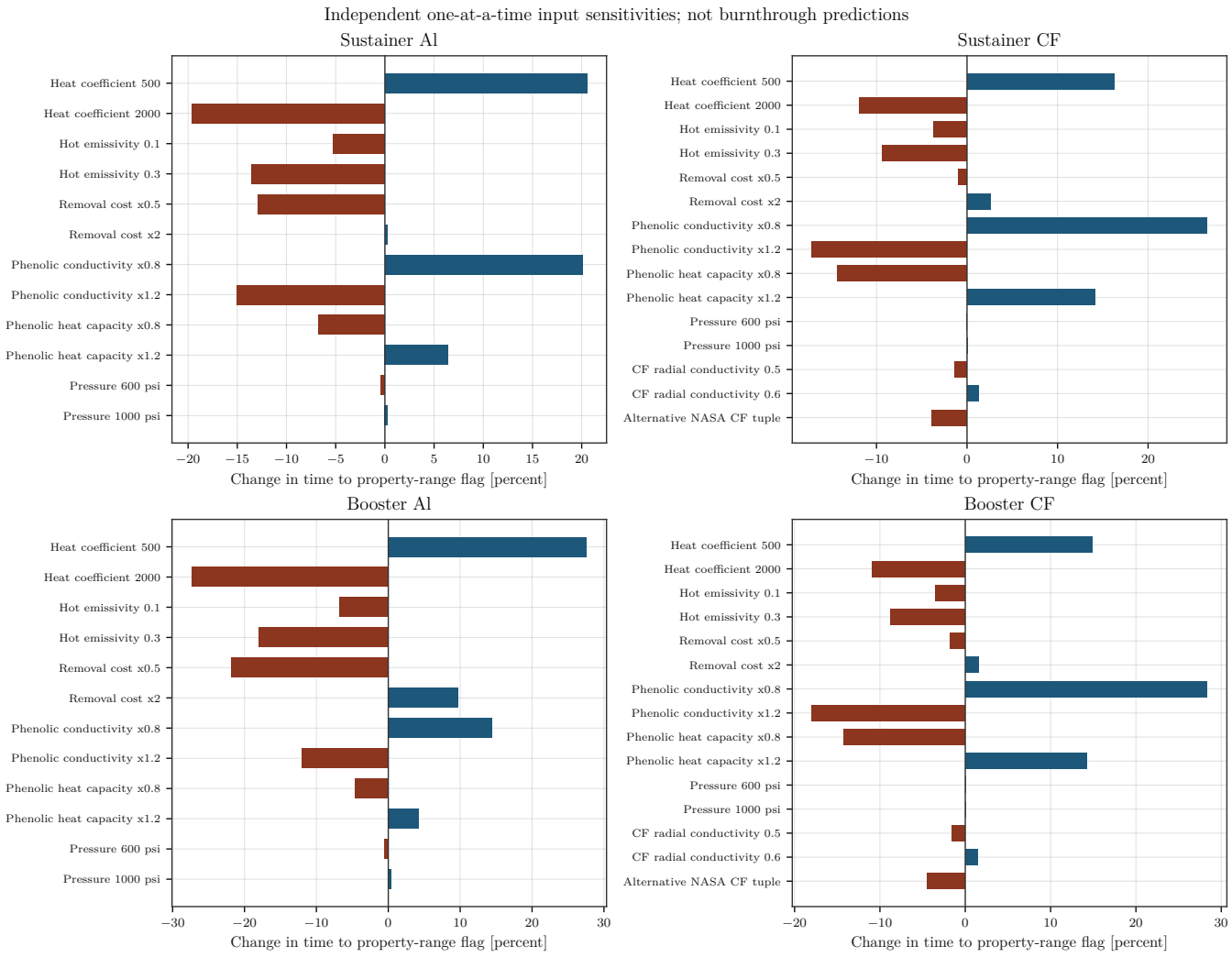


Figure 9. Changes in the duration of prescribed constant heating before the provisional property-range flag. Red bars reach the flag sooner, blue bars later. These bars are not safe firing durations or burnthrough margins.

10.4 Numerical resolution of the sensitivity screen

Ten stronger-heating, lower-phenolic-conductivity and alternative-CF scenarios were repeated with 160 radial cells and half the step ceiling. Their property-scope stopping times changed by at most 1.39 percent. This checks selected scenarios, not every bar. Changes comparable to percent-level discretization error are not resolved physical differences.

Remaining thickness was checked separately. The initial 80-to-160-cell comparisons exceeded the stated thickness tolerance in three scenarios, despite agreement in stopping time. Those cases received another refinement rather than being declared converged from timing alone.

Case	Perturbation	Timing change [%]	Thickness change [mm]	Target [mm]
Sustainer Al	Higher heating	3.50	0.3870	0.0476
Booster CF	Higher heating	0.72	0.0794	0.0635
Booster CF	Lower conductivity	0.48	0.0000	0.0635

This table compares 160 with 320 cells, with a further halving of the time step. The remaining-thickness failures in this comparison prompted the smaller-step investigation below.

10.5 Resolution of the stronger-heating coupling error

The two unresolved doubled-heating cases were recomputed at 320 cells with time steps of 0.0001953125 and 0.00009765625 s, and at 640 cells with 0.000048828125 s. The joint comparison scales the time step with radial

pitch squared. All reported time-only and joint comparisons now meet the five-percent timing target and the retained-thickness target of the larger of 0.02 mm and one percent of initial liner thickness.

Case	640-cell scope stop [s]	Retained [mm]	Joint timing change [%]	Thickness change [mm]
Sustainer Al	39.8790	1.9422	0.100	0.0223
Booster CF	29.7066	4.4648	0.102	0.0198

These finer independent values supersede the coarse stronger-heating recession estimates. They resolve the tested numerical comparisons, not the unknown char kinetics or actual heat load. A closed mass ledger alone did not reveal the earlier time-coupling error. The retained FE thickness remains a whole-row geometric quantity, not a measured intact liner thickness.

10.6 Retained-char heat-capacity diagnostic

Twelve paired independent checks compare the existing full-cell capacity with an optional capacity-only reduction for consumed surface char, at 80, 160 and 320 phenolic cells. Each pair has the same mesh, time schedule and seven-second pulse. The variant changes heat capacity only: conductivity, geometry and source laws remain fixed. It is not a conservative enthalpy-removal model, a validated alternative, or a rigorous uncertainty bound.

Case (320 cells)	Peak change [K]	Scope-time change [%]	Thickness change [mm]
Sustainer Al	-0.265	No flag	0.0000
Sustainer CF	Censored	0.530	0.0000
Booster Al	-0.348	No flag	0.0000
Booster CF	-0.453	No flag	0.0000

At 320 cells the complete-history inner casing peaks change by less than 0.46 K, while the sustainer CF scope time changes by 0.53 percent; no paired retained-thickness change occurs. The maximum consumed char still represented in a live FE row is about 0.143 percent of initial liner mass. At 80 cells, the sustainer CF flag shift is 3.74 percent, so this approximation is more visible on the coarser mesh. These conditional numerical effects do not bound unmodelled cracks, spallation or actual material behavior.

10.7 Native prescribed-temperature reaction check

The native user material was held at uniform prescribed phenolic temperatures of 300, 600 and 1000 degrees Celsius. Conversion at 0.1, 1 and 5 s matched the analytical first-order Arrhenius solution within 1.7e-08 absolute conversion. This verifies the implementation under controlled temperature; it does not validate reaction constants against the actual liner. No surface-consumption controller was used in this fixture.

10.8 Rejected refined step and restart correction

The first 80-row sustainer Al run stopped at an attempted 4.175 s: its energy residual was 10.420 percent, above the unchanged five-percent limit. Its last accepted state was 4.1625 s. There was no internal substep cutback. Investigation found that an earlier removed layer returned after a multiframe restart because the controller reissued only newly pending deactivations. The rejected state is excluded from all accepted results.

ANSYS documents that multiframe restart files do not save element birth/death commands; these must be reissued after restart. The corrected controller replays the entire removed prefix, checks the native live-element count before every solve, and stores two hash-verified accepted native restart generations. See the ANSYS 2026 R1 Basic Analysis Guide, multiframe restart limitations.

The failed runtime and original queued packages are preserved. Fresh sibling packages use identical physical inputs and audit tolerances. The old native restart ring lacked the recent accepted state, so a failed native state was not combined with an earlier controller checkpoint.

A manufactured transient regression reproduced the old resurrection defect for all four cases. Full deactivation replay retained the correct live-element counts and exactly matched uninterrupted nodal temperatures at all sixteen checkpoints. The uninterrupted reference remains in the solution processor until its load steps finish; two earlier fixture drafts incorrectly re-entered a new solution and are preserved as rejected diagnostics. This regression verifies restart topology, not ablation physics.

10.9 Cooling withdrawn after an internal-substep audit gap

The corrected restart run completed seven seconds of heating with one native substep per controller interval. At the first cooling interval, ANSYS cut back the requested 0.25 s step and solved at 7.125, approximately 7.19483, and 7.25 s. The old controller combined the final internal-substep capacity and gas-rate exports with the conversion increment over the whole controller interval. Passing its numerical threshold did not establish a consistent interval energy balance.

The entire subsequent cooling chain is therefore withdrawn from accepted results, and the original run was paused and preserved. The heating prefix through 7 s is retained after inspection of every native load-step entry. Only the final result of each controller interval was saved, so the missing intermediate cooling energy terms cannot be reconstructed faithfully after the run.

Fresh packages export native load-step and substep identity. Acceptance now requires exactly one native substep for each controller interval. The first second of cooling uses 0.0625 s intervals before returning to 0.25 s; any further internal cutback stops acceptance and preserves the preceding accepted restart. Separate default and tighter heat-flow convergence controls use identical schedules through 20 s. Their completed comparison is reported below; the acceptance gates were not relaxed.

A native read-only result probe returned substep 1 at 7 s and substep 3 at 7.25 s; the new acceptance function admits the former and rejects the latter. Retrospective log inspection found only single-substep intervals in all four published coarse baselines and the earlier restart/pulse controls. Their published acceptance is unaffected.

10.10 Reaction-temperature convention and nonlinear tolerance

The intended temporal midpoint is $(T_{n+1}+T_n)/2$. The inherited routine instead added half the temperature increment to the ANSYS temperature argument. The ANSYS UserMatTh interface supplies the current temperature and its increment. Accepted native-temperature replay identifies the actual evaluation as $T_{n+1} + \Delta T/2$, rather than the intended $T_{n+1} - \Delta T/2$. This affects both conversion and the gas sensible-energy sink.

Across the completed 20 s control, independent integration driven by every saved native temperature reproduces conversion with maximum absolute error 2.89e-08 for the inherited expression. The intended midpoint expression differs by up to 0.01819. This isolates the interface convention from differences in the temperature solver.

The earlier isothermal reaction checks could not detect this error: the temperature increment was zero. Their successful constant-temperature results remain valid. The historical native curves elsewhere in this report retain their actual inherited routine and source hashes; they must not be interpreted as verification of the intended midpoint implementation. Earlier independent refinement and sensitivity calculations already use the intended midpoint.

Six additional native prescribed-temperature checks include increasing, decreasing and constant temperatures, each over one 0.01 s step. The inherited library matches the shifted expression; the separately rebuilt candidate matches the intended midpoint. Neither check calibrates reaction constants against a real liner.

Start [°C]	End [°C]	Intended conversion	Inherited native	Corrected native
300	600	7.83092e-05	0.00178077	7.83092e-05
600	1000	0.00252853	0.0176384	0.00252853
1000	600	0.00252853	3.54829e-05	0.00252853
600	300	7.83092e-05	4.09532e-08	7.83092e-05
600	600	0.000488519	0.000488519	0.000488519
1000	1000	0.00779275	0.00779275	0.00779275

The correction changes only the sign of half the increment in the material routine. A new Intel Fortran x64 library was built in an isolated study folder; archived failed and historical runtimes are unchanged. Fresh packages must pass native audits and independent comparison before their thermal results supersede the inherited histories.

The corrected 80/20-cell sustainer Al motor pilot completed eight audited intervals through 0.1 s. Its largest native/independent temperature-profile difference was 0.00124 K. This permits longer corrected verification runs; it does not establish convergence of recession or a complete pulse/cooling history.

Both separate default/tighter nonlinear-tolerance controls completed 20 s with 624 accepted single-substep intervals each. Persistent layer removal, cooling boundary transfer and native balance checks passed. Their maximum live-temperature difference was 6.986 K. Tightening the heat-flow tolerance is a separate numerical correction; it cannot repair the reaction-temperature expression.

Native tolerance	Versus intended midpoint [K]	Versus inherited expression [K]
Default	8.638	6.757
Tighter	5.173	0.229

The last column repeats the independent assembly with the inherited shifted temperature solely to diagnose the old native implementation. It does not replace the intended midpoint model, and agreement is not physical validation. Exact row-event scheduling remains a separate check.

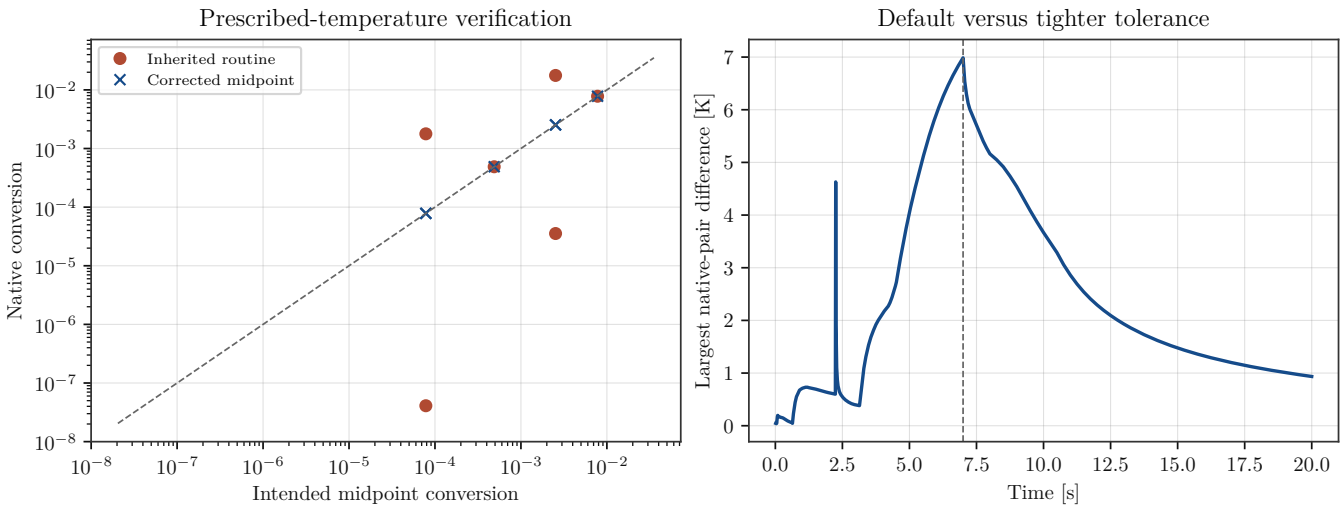


Figure 10. Separate software checks of the reaction-time convention and nonlinear solver tolerance. The dashed diagonal is exact midpoint agreement; the vertical line marks the imposed seven-second heating pulse. These are numerical verification results, not measured motor behaviour.

10.11 Refined native pulse cross-checks

These comparisons use the independent assembly at the same 80/20 radial mesh and coupling schedule as the native pulse cases. Only accepted native checkpoint times are compared; a running case has no predicted ANSYS future values. Layer-removal scheduling and the next hot-face transfer are checked separately.

Case	Through [s]	Max profile [K]	Max conversion	Transfers passed
Sustainer Al	127.000	0.333	0.00027	1
Sustainer CF	34.000	0.333	0.00027	1
Booster Al	300.000	0.220	0.00016	1
Booster CF	300.000	0.220	0.00016	1

All displayed comparisons meet the same-mesh profile, conversion, mass and row-count checks.

These checks do not establish fine-mesh convergence of the native stopping time. In particular, the independent sustainer CF pulse needed further refinement of its provisional property-flag time.

This table uses fresh corrected-midpoint native runs. The earlier seven-second historical comparison and withdrawn cooling chain are excluded. Agreement now checks the intended midpoint reaction model, persistent layer deactivation and the strict single-substep cooling transition together.

Case	Row	Native scheduled [s]	Independent [s]	Difference [s]
Sustainer Al	1	4.1625	4.1625	0.00000
Sustainer CF	1	4.1625	4.1625	0.00000
Booster Al	1	4.6000	4.6000	0.00000
Booster CF	1	4.6000	4.6000	0.00000

These are individual row-removal events, not liner-exhaustion times. The native event is bracketed by its previous accepted intact state and the scheduled state; geometric deactivation occurs in the next solve.

10.12 Variation within the uniform sector

Every accepted checkpoint in the completed corrected primary cases was checked for differences among the nine nodes at each radial station. The table reports the largest axial/angular spread at a common radius, using only nodes belonging to elements active during that solve. This quantifies information suppressed by radial averaging; it does not establish uniformity in a real motor or bound unmodelled end effects.

Case	Through [s]	Phenolic range [K]	Casing range [K]
Sustainer Al	127.00	1.19e-08	1.13e-12
Sustainer CF	34.00	0.00653	0.013
Booster Al	300.00	1.07e-08	4.55e-13
Booster CF	300.00	0.00575	0.0104

10.13 Native spatial and time-resolution program

Thirteen additional corrected-material native packages have been prepared. For each motor/casing pair, the first comparison keeps 80 phenolic and 20 casing rows while reducing the heating and cooling coupling steps by four. The next uses 160/40 rows at the same refined steps, separating spatial from time error. A 320/80-row case then halves those steps. Sustainer CF additionally uses 640/160 rows because the independent 160-to-320-cell scope-time comparison failed its target.

The 80-row time check and 160-row spatial check use 0.003125 s heating, 0.015625 s during the first cooling second, and 0.0625 s thereafter. The 320-row case halves these intervals; the 640-row case halves them again. All use the same physical inputs, corrected library, tighter nonlinear tolerance and unchanged audit gates. Each package preserves the primary cooling horizon and stops at the provisional material flag.

Native comparisons require casing peak changes below 5 K for completed histories, scope-time changes below five percent for censored histories, and retained-thickness changes no larger than the greater of 0.02 mm or one percent of the initial liner. A scope-limited calculation cannot supply a later casing peak. Preparation is not an accepted result.

At this edition no completed native refinement pair is available. The independent refinement results below remain distinct.

10.14 Sensitivity to the row-removal cutoff

The removal threshold is a numerical choice: a row is deactivated when residual solid mass is less than 0.1 percent of its initial virgin mass. Twenty-four paired independent runs tested 0.01, 0.1 and 1 percent cutoffs for all four cases at 160/40 and 320/80 phenolic/casing cells. Physical laws and time settings were fixed within each pair. These are numerical-tolerance perturbations, not material uncertainty.

Case (320 cells)	Cutoff [%]	Inner peak change [K]	Scope-time change [%]	Retained change [mm]
Sustainer Al	0.01	2.399	–	0.0149
Sustainer Al	1	-3.812	–	-0.0298
Sustainer CF	0.01	Censored	-7.211	0.0149
Sustainer CF	1	Censored	16.119	-0.0298
Booster Al	0.01	0.973	–	0.0000
Booster Al	1	-2.636	–	-0.0397
Booster CF	0.01	1.406	–	0.0000
Booster CF	1	-3.716	–	-0.0397

Changes are relative to the 0.1-percent calculation on the same mesh. At 320 cells, complete-history casing peaks shift by as much as 3.812 K. Sustainer CF scope timing changes by about -7.21 to +16.12 percent, exceeding the five-percent timing target. This sensitivity is separate from radial/time convergence at a fixed cutoff. The queued native cases retain their sealed 0.1-percent rule; their reported times remain conditional on that choice.

The small expelled residual mass does not make this timing effect negligible. A nearly consumed row can remain thermally active while the residual virgin fraction continues reacting, delaying transfer of the hot boundary to the next row. This explains why a mass-ledger check alone does not settle removal discretization. No real-motor lifetime follows from these cutoff-dependent states.

The chosen equations also explain why a zero cutoff is not a physically validated target. Char consumption cannot exceed the available char, so $m_{\mathrm{solid}}/V \geq \rho_v(1-\alpha)$. The first-order conversion law leaves $1-\alpha = (1-\alpha_0)\exp(-\int k\,dt) > 0$ at finite integrated reaction exposure. Thus exact zero residual is not reached in finite time in the continuous equations. The finite cutoff approximates ejection or loss of continuity of the residual solid, for which this study has no calibration. Tighter cutoffs test numerical/model dependence; they do not automatically improve physical accuracy.

Additional fixed-640-cell sustainer CF checks at fractional cutoffs 1e-7 and 1e-8 change the scope time by 1.369 and 1.514 percent relative to the preceding decade. The 1e-8 case stops at 24.3906 s with 4.7551 mm retained. These smaller successive changes do not erase the difference from the original cutoff or identify a physically correct residual-ejection rule.

Additional independent calculations now compare successive fractional cutoffs of 1e-4, 1e-5 and 1e-6 at 320 and 640 cells. These tests concern the numerical equations, not physical validation of the removal law.

Case	Final comparison	Peak change [K]	Scope change [%]	Retained change [mm]	Pass
Sustainer CF	Cutoff	–	-2.606	0.0074	Yes
Sustainer CF	Mesh/time	–	-2.488	0.0000	Yes
Sustainer Al	Cutoff	1.078	–	0.0074	Yes
Sustainer Al	Mesh/time	0.868	–	0.0000	Yes
Booster Al	Cutoff	0.490	–	0.0000	Yes
Booster Al	Mesh/time	0.522	–	0.0000	Yes
Booster CF	Cutoff	0.705	–	0.0000	Yes
Booster CF	Mesh/time	0.745	–	0.0000	Yes

Passing a successive-refinement target does not prove a zero-cutoff limit or remove the difference from the original 0.1-percent choice. Native confirmation and interpretation of the tighter rule remain separate work; no queued sealed configuration is silently replaced.

A separate native 80/20-row sustainer CF control with fractional cutoff 1e-6 has been prepared but has not completed both verification gates. Its time schedule matches the queued 80-row quarter-step default-cutoff case, allowing a controlled later native comparison. Neither sealed input is changed in place.

10.15 Combined-input pulse scenarios

Two joint engineering perturbations were calculated for all four configurations, separately from ANSYS. The first doubles the heat-transfer coefficient to 2000 in SI units, raises phenolic conductivity by 20 percent and lowers its heat capacity by 20 percent. The second doubles the same coefficient, lowers phenolic conductivity by 20 percent and halves the effective char-removal cost to 17.5 MJ/kg. Every change comes from the earlier one-at-a-time engineering screen; their combination is neither a measured uncertainty interval nor a demonstrated worst case.

All cases use a hypothetical seven-second heating pulse with the primary ambient-cooling assumptions and property-scope stops. Eight scenarios were each repeated with 80/20, 160/40 and 320/80 phenolic/casing cells. Heating steps scale with radial pitch squared: 0.003125, 0.00078125 and 0.0001953125 s. Later cooling steps are respectively 0.0625, 0.015625 and 0.00390625 s. Intermediate time and mesh effects remain coupled in this additional input screen; the separate native resolution program tests those effects independently.

All eight final comparisons meet the stated targets. Complete-history casing maxima change by at most 0.855 K; scope-time changes are at most 0.259 percent. Retained-thickness checks pass separately. These numerical results do not calibrate the joint physical assumptions.

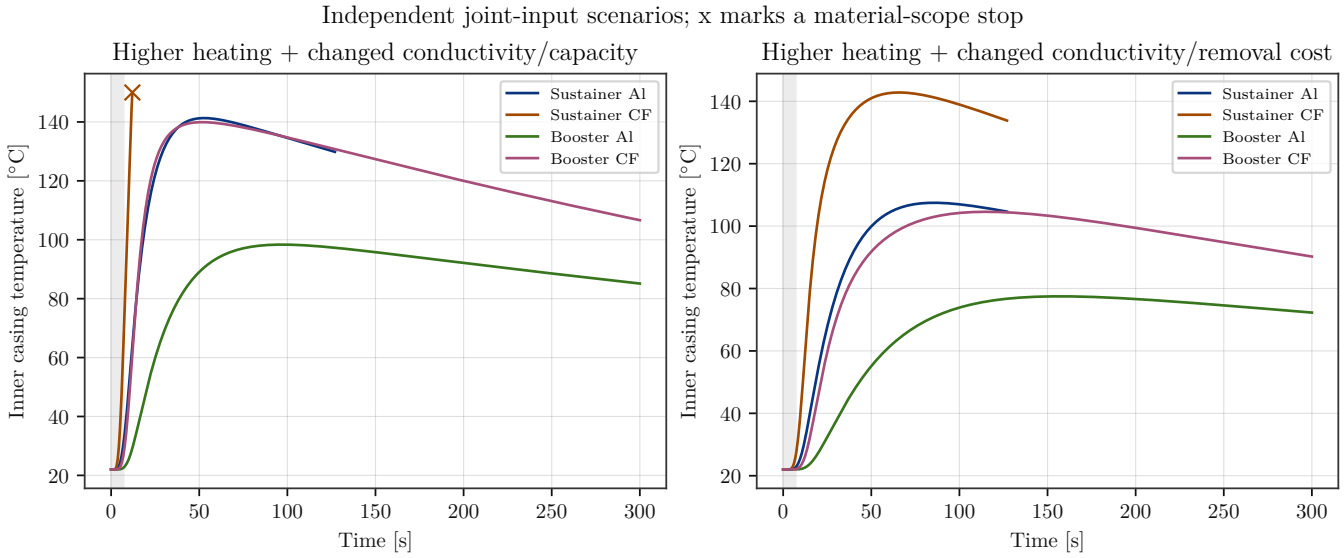


Figure 11. Independent 320-row combined scenarios. The shaded interval is imposed heating. Scope-limited curves do not predict a later casing peak or structural failure.

Case	Joint scenario	Through [s]	Inner max [°C]	Retained [mm]
Sustainer Al	Transport	127.000	141.31	4.5988
Sustainer Al	Consumption	127.000	107.48	4.3607
Sustainer CF	Transport	12.047	150.01	4.5988
Sustainer CF	Consumption	127.000	142.83	4.3607
Booster Al	Transport	300.000	98.36	6.1912
Booster Al	Consumption	300.000	77.50	5.9730
Booster CF	Transport	300.000	139.93	6.1912
Booster CF	Consumption	300.000	104.59	5.9730

Changing several assumptions can change both heat admitted to the wall and energy removed by gas generation and char consumption. A larger imposed heat-transfer coefficient therefore need not produce a hotter casing when the other closure parameters also change. This is sensitivity of the chosen equations, not evidence that changing a real liner would improve safety. No scenario here establishes a real-motor burnthrough time.

Four separate bounded native controls have been prepared for higher heating, the published alternative CF tuple and two combined cases. They stop at 20 s or an earlier material flag and test implementation against the same-mesh independent model. Their short horizons cannot establish later cooling maxima.

At this snapshot, 0 of those four controls have completed both native audit and independent comparison. Prepared or running controls supply no completed native confirmation.

11 Independent pulse and cooling investigation

These curves are independent radial calculations, not ANSYS output. Each uses seven seconds of the prescribed hot-gas boundary followed by ambient gas with a heat-transfer coefficient of 8 in SI units, zero hot radiation and zero char consumption. Pyrolysis and conduction continue during heat soak. The pulse duration and cooling conditions are assumed, not measured firing or tailoff histories.

The sustainer Al and both booster references use 320 phenolic and 80 casing cells. Sustainer CF required 640 and 160 cells to bring its property-flag timing change below five percent. Its curve stops at the provisional 150-degree casing flag, so its later peak is unknown within this material model. The other curves end after their sampled casing peaks.

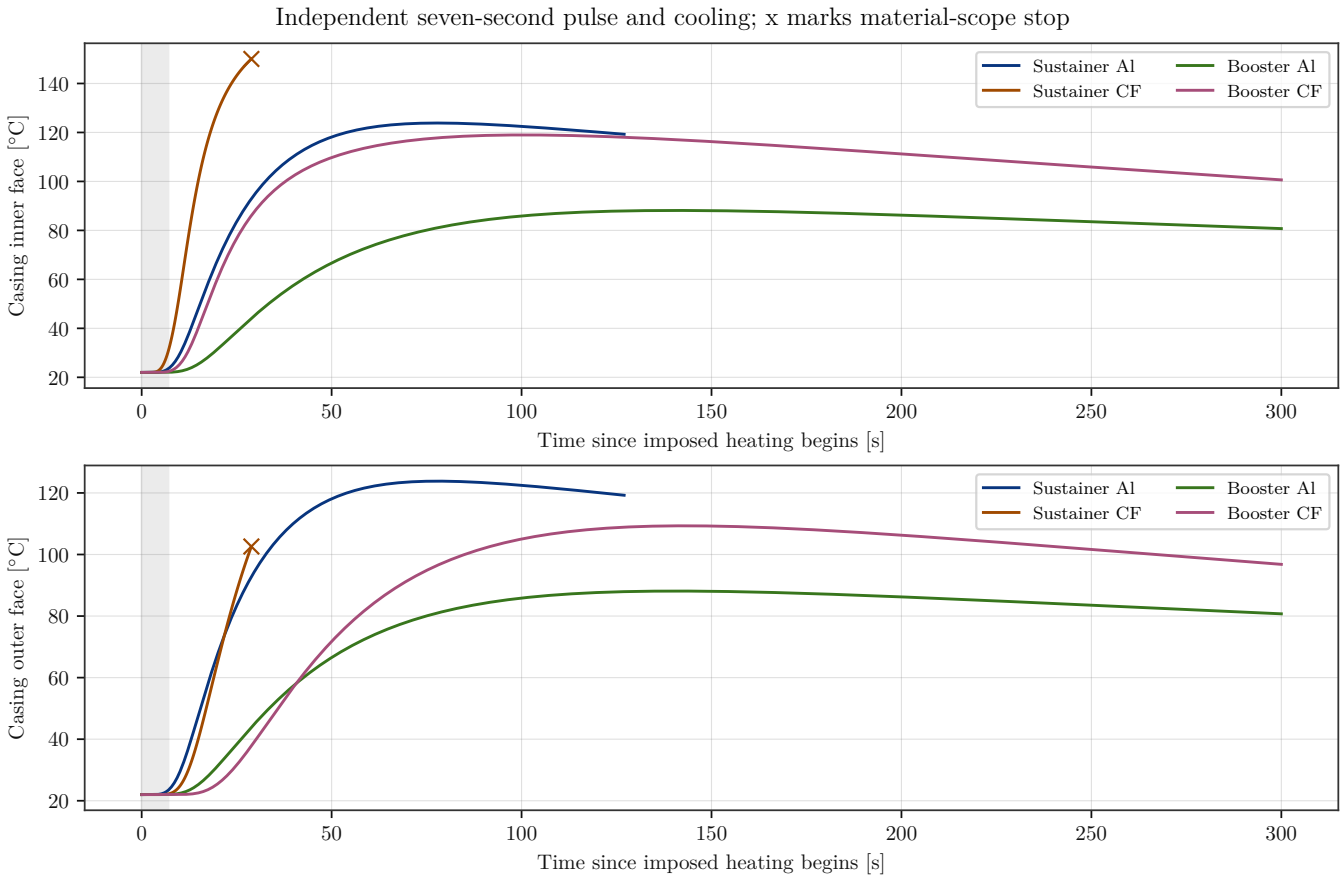


Figure 12. Independent refined temperature histories. The shaded interval is imposed heating; curves beyond it describe assumed cooling. A scope stop is not a predicted casing failure.

Case	Through [s]	Inner max [°C]	Outer max [°C]	Retained mesh [mm]
Sustainer Al	127.00	123.85	123.83	4.7179
Sustainer CF	28.88	150.01	102.60	4.7179
Booster Al	300.00	88.14	88.12	6.3103
Booster CF	300.00	118.99	109.32	6.3103

For the complete 127-second histories, the last 160-to-320-cell joint refinement changes sampled casing maxima by at most 1.7 K and remaining mesh thickness by at most 0.015 mm. Sustainer CF instead compares its censored stopping time; the 320-to-640-cell change is about two percent. These checks do not establish convergence of every hot-face temperature or a future exhaustion event.

Both booster cases still had a rising casing temperature at 127 s. Independent extension to 300 s captured their later peaks, so fresh native booster packages were prepared with that longer cooling horizon. The original unstarted packages are preserved. Native confirmation of the refined pulse/cooling histories remains pending.

11.1 Exploratory twenty- and sixty-second pulses

These durations are assumed scenarios, not firing-time estimates. Each calculation stops at its provisional casing-property flag or continues for 300 s after heating. A stop during heating means the prescribed pulse was not completed within the material model; a stop during cooling identifies delayed heat soak. None reached phenolic exhaustion within the accepted model scope.

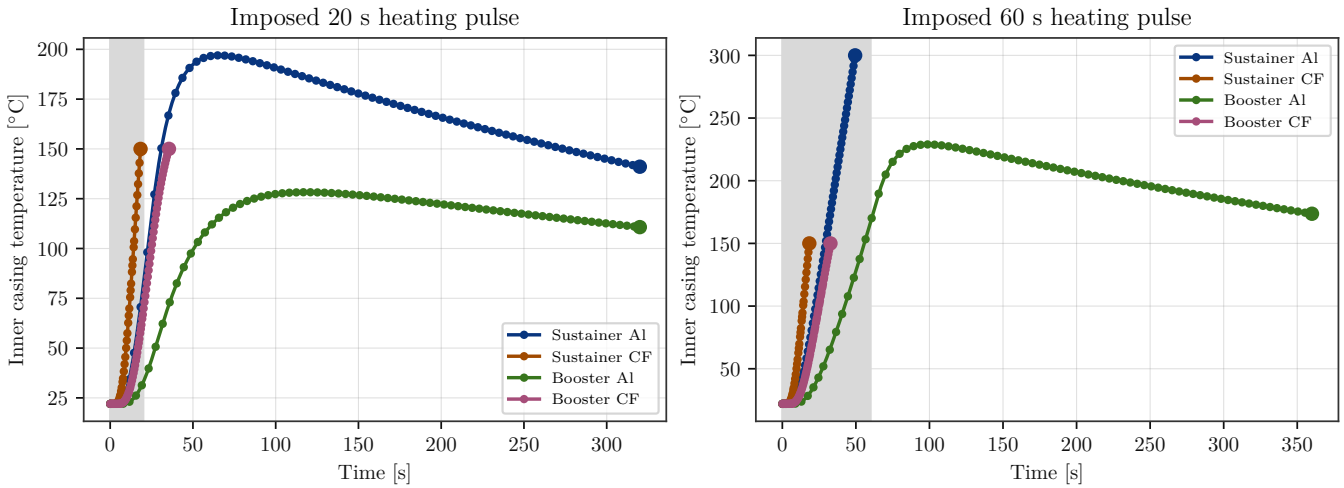


Figure 13. Independent 320-cell longer-pulse scenarios; gray shading marks prescribed heating. Curves end at a scope flag or at the scenario horizon.

Case	Pulse [s]	Through [s]	Stop reason	Inner max [°C]	Retained [mm]
Sustainer Al	20	320.000	Horizon	196.96	4.3458
Sustainer Al	60	49.486	During heating	300.00	3.0361
Sustainer CF	20	18.344	During heating	150.00	4.4053
Sustainer CF	60	18.344	During heating	150.00	4.4053
Booster Al	20	320.000	Horizon	128.25	5.9531
Booster Al	60	360.000	Horizon	228.96	4.1870
Booster CF	20	35.500	During cooling	150.03	5.9531
Booster CF	60	32.709	During heating	150.00	5.4570

All eight 160-to-320-cell comparisons passed their numerical targets. Censored flag times changed by at most 1.05 percent; complete-history casing maxima changed by less than 0.97 K, and retained thickness changed by at most 0.020 mm. Sustainer CF reaches the same scope stop before either longer pulse ends, so those two rows describe the same partial heating history. These independent scenarios still require native confirmation of influential conditions and measured heating/material data for a physical lifetime claim.

12 Next results

Complete the corrected native refined pulse/cooling cases and the remaining native resolution and influential-input checks. Compare persistent first-row removal, cooling, temperature profiles and mass loss against the independent assembly, within the stated material scope. Report whole-cell recession separately from the 98 percent pyrolysis front. Preserve rejected diagnostics with their reasons rather than silently treating them as successful predictions.

A native one-second heating pulse followed by five cooling steps passed the boundary-transfer and balance checks. This verifies the prescribed switch to ambient gas and zero char consumption; it does not measure real tailoff.

A defensible final answer may be a conditional “not reached through the accepted exposure” result with remaining thickness, rather than a finite lifetime. Neither missing wall-exposure data nor uncalibrated char kinetics can be recovered from solver convergence. The study cannot establish real motor lifetime without supported wall heating and recession evidence.

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